

Forecast Uncertainty in Battery Dispatch: Sizing Invariance Persists Through European Price Spikes

Julian Quick

Lawrence Berkeley National Laboratory

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Abstract

Tesla, Fluence, and Wäertsilä run battery dispatch on machine-learning-driven stochastic optimization. NREL, DTU, and most academic hybrid-power-plant sizing tools (**REopt**, **hydesign**, **PyPSA**) embed deterministic LP inner dispatch. Are sizing tools wrong? We test the question on a controlled benchmark with rainflow degradation, then on three years of Danish DK1 day-ahead prices spanning a typical, energy-crisis, and recovery year. Across all four tested settings the optimal battery energy capacity b_E^* is invariant under deterministic vs. multi-lag-ensemble dispatch, despite stochastic dispatch yielding 9–34% realised-NPV uplift at the optimum. We sketch a theoretical condition under which this invariance is principled (uniform-in-capacity revenue lift), test it empirically, and find that the prediction — that multi-day spectral content should break invariance — is falsified on real data. The empirical answer is broader than the theory predicted: sizing-tool fidelity is structurally robust across regimes including the 2022 European energy crisis, with revenue uplift translating roughly uniformly across capacity. We release a practitioner-runnable diagnostic (b_{sat}^ϵ overlap test) and the benchmark (**battery_gym**) for replication.

1 Introduction

A battery operator’s two coupled questions are *what to build* (capacity) and *how to operate it* (dispatch under uncertain prices and renewables). Industry decomposes them: capacity is a one-shot investment, dispatch is continuous re-optimization. The two communities of practice make different modelling choices. Tesla Autobidder (> 100 GWh dispatched) explicitly combines “classical statistics, machine learning and numerical optimization” (Tesla, 2024); Fluence Mosaic uses “stochastic optimization models” on ML price forecasts (Fluence, 2024); Wäertsilä GEMS Pulse markets predictive analytics for revenue resilience (Wäertsilä, 2025). These deployments together cover tens of gigawatts. Meanwhile, academic hybrid-power-plant sizing tools (**hydesign** (Murcia León et al., 2024), **REopt** (NREL, 2025), **PyPSA** capacity-expansion variants) approximate inner dispatch as deterministic LP fed point-forecast prices. The outer surrogate optimizer evaluates this LP 10^3 – 10^4 times; making it stochastic would cost $100\times$ – $1000\times$ more compute per evaluation.

Are the sizing tools structurally wrong, or is the architectural divergence principled? This paper answers that question empirically.

We construct a controlled hourly-arbitrage benchmark with battery rainflow degradation and replacement scheduling, run a 2×2 factorial {linear, quadratic} cycling cost $\times$ {single, multi-lag $K=4$ ensemble} forecast-handling, and report sizing $\text{argmax} + \text{NPV}$ at optimum across (a) a synthetic diurnal AR(1) regime and (b) Danish DK1 day-ahead prices for 2021 (typical), 2022 (European energy crisis with multiple >500/MWh spike days), and 2023 (recovery, occasional

−440/MWh negative-price hours). Pre-registered protocol locks all design choices before data touch (commits 2ce93ae, ebb035c, see PREREGISTRATION_AMEND_01.md). The contribution:

1. **Operational lift confirms the literature.** Multi-lag ensemble dispatch yields +9–34% realized lifetime NPV over single-forecast dispatch under both linear and quadratic cycling cost. Consistent with [Cao et al. \(2020\)](#), [Krishnamurthy et al. \(2018\)](#).
2. **Sizing argmax is invariant** across all four price regimes tested (synthetic + DK1 $\times$ 3 years), all four policies, and all noise levels in the synthetic sweep. The lift is approximately constant in b_E .
3. **Theoretical condition partially derived, empirically falsified in its prediction.** A proposition relating revenue plateau capacity $b_{\text{sat}}(\pi)$ to a Wiener-style forecast-resolvable timescale $\tau_{\text{res}}(\pi)$ predicts that multi-day spectral content should break invariance. DK1 has strong 168h (weekly) and 336h (fortnightly) spectral peaks; nevertheless invariance holds. The empirical answer is broader than the prediction: invariance does not require single-timescale price processes.
4. **Practitioner-runnable diagnostic.** The b_{sat}^ϵ overlap test is a regime classifier any operator can run on their own price data without committing to the proposition.

The result has practical implications: deterministic-LP-inner-dispatch in academic sizing tools appears to be empirically robust, including under spike-heavy regimes, on the dimensions tested. Operational stochastic-dispatch literature is independently justified by the realized-NPV uplift.

2 Setup

2.1 Environment and dispatch

A single-battery hourly-arbitrage MDP. State is continuous SoC; the exogenous signal is an hourly price trace. Actions $a \in [-b_P, +b_P]$ where positive = discharge. SoC dynamics $\text{SoC}_{t+1} = \text{SoC}_t - a_t$ (efficiency $\eta = 1$ throughout). Per-step reward $r_t = p_t a_t - \alpha a_t^2$; the quadratic penalty is a convex surrogate for cycle depth, consistent with the convex-cycle relaxation of [Shi et al. \(2017\)](#).

Lifetime degradation is computed post-hoc via rainflow cycle counting on the SoC trace using the ASTM E1049 algorithm and the stress function $f(\delta) = (k_1 \delta^{k_2} + k_3)^{-1}$ ([Xu, 2018](#)). Replacements are scheduled when cumulative D consumes a 20% loss-of-health budget.

2.2 Price processes

Synthetic. Diurnal AR(1) per-hour price as in `price.signal.synth.diurnal`: mean 50, two diurnal harmonics (24h, 12h), AR(1) noise $\rho = 0.7$, $\sigma = 8$. Marginal mean 48.5, std 28.4.

DK1 day-ahead. Energinet ([Energinet, 2024](#)) hourly DA spot prices for the DK1 area (West Denmark / Jutland), 2021/2022/2023. Statistics in Table 1. The 2022 prices are an outlier within the post-2015 record: mean 219/MWh (vs 50–90 typical), max 871/MWh (vs 200 typical), driven by gas-supply uncertainty post-Ukraine. 2023 has occasional deeply-negative prices (−440/MWh) from oversupply during low-demand intervals.

Table 1: DK1 day-ahead hourly statistics. Persistence($t - 24\text{h}$) residuals are computed on the realized trace.

Year	Mean	Std	Min	Max	Pers. residual std	Excess kurtosis
2021	88.1	64.8	-43.9	620.0	46.4	8.7
2022	219.0	145.4	-19.0	871.0	84.8	3.3
2023	86.8	48.8	-440.1	524.3	43.3	7.3

2.3 Forecasts and policies

Single forecast. Persistence at $t - 24\text{h}$: $\hat{p}_t = p_{t-24}$. Standard baseline in price-forecasting literature; empirically biased and fat-tailed (Table 1).

Multi-lag ensemble ($K=4$). Persistence at lags $\{24, 48, 168, 336\}$ hours, averaged. Each member of the ensemble is a different hypothesis about which past pattern best predicts the present; averaging cancels lag-specific biases (cf. [Hong et al. \(2014\)](#), [Weron \(2014\)](#)).

2×2 policy factorial. {linear LP via `scipy.linprog` HiGHS, quadratic QP via `cvxpy` CLARA-BEL} $\times$ {single, ensemble}. Both costs use the same cycling-cost coefficient $\alpha = 0.005$ (QP) or $\mu = 5.0$ EUR/MWh (LP). The 2×2 separates cost-form effects from forecast-handling effects per a referee’s pre-publication critique. We run $b_E \in \{0.5, 1, 2, 4, 8, 16, 24, 32, 48, 64, 96, 128, 192, 256\}$ MWh; $b_P = 1$ MW fixed. Year-long horizons are split into 8-week chunks for solver tractability; SoC carries forward across chunks.

2.4 Sizing model

Lifetime NPV = $R_{\text{annual}} \cdot \sum_{y=0}^{14} (1.07)^{-y} - C_E b_E - C_P b_P - \text{replacement CAPEX}$. $C_E = 100\text{k/MWh}$, $C_P = 75\text{k/MW}$ (PCS overhead), 15-yr lifetime, 7% discount. Replacements occur at evenly-spaced years when annual degradation $\times$ years exceeds the LoH budget; only the energy component is re-paid (PCS reused).

3 Theory

3.1 Setup

Let $R(b, \pi)$ be the expected lifetime arbitrage revenue under capacity b and policy π . CAPEX is linear in b . Sizing $\arg\max b^*(\pi) = \arg\max_b R(b, \pi) - C_E b - C_P b_P - \text{repl}(b, \pi)$.

Define the lift $\Delta(b) = R(b, \pi_{\text{ens}}) - R(b, \pi_{\text{single}})$. By the first-order condition, the $\arg\max b^*$ satisfies $R'(b^*, \pi) = C_E + \text{repl}'(b^*, \pi)$. So b^* is invariant across π when $\Delta'(b^*) = 0$ (assuming replacement-cost derivative agrees in the relevant neighborhood, which we verify empirically).

3.2 Sufficient mechanistic condition (proposition)

Consider a price process $p(t)$ with spectral density $S_p(\omega)$. A pure tone at frequency ω_0 , period $\tau_0 = 2\pi/\omega_0$, has dispatch-relevant capacity threshold $b_E^{\text{cycle}}(\tau_0) = b_P \tau_0 / 2$. Capacity past this threshold contributes nothing to one-cycle revenue.

Define $q(\omega, \pi) \in [0, 1]$ as the fraction of price variance at frequency ω that policy π resolves from its forecast input. For a Gaussian-noise single forecast with marginal noise variance σ_f^2 and AR(1)

Table 2: NPV at the optimal b_E^* for each (year, policy). b_E^* is invariant across policies within a year; lift is the relative NPV uplift of QP-ensemble over QP-single at b_E^* .

Year	b_E^* (MWh)	NPV LP-single	NPV LP-ens	NPV QP-single	NPV QP-ens	
2021	4	247k	392k	310k	416k	(+34%)
2022	8	1.53M	1.86M	1.59M	1.91M	(+20%)
2023	4	386k	548k	462k	575k	(+25%)

autocorrelation ρ_f ,

$$q(\omega, \pi_{\text{single}}) \approx \frac{S_p(\omega)}{S_p(\omega) + \sigma_f^2 / (1 - \rho_f^2)}$$

(Wiener-style SNR per frequency). Long timescales — low frequencies — have signal power concentrated in fewer modes and are most sensitive to the noise floor.

Let $\tau_{\text{res}}(\pi) = 2\pi/\omega_{\text{cut}}(\pi)$ where ω_{cut} is the lowest frequency at which q drops below threshold. Then we claim $b_{\text{sat}}(\pi) \approx b_P \tau_{\text{res}}(\pi)/2$. Argmax invariance follows when $\tau_{\text{res}}(\pi_1) = \tau_{\text{res}}(\pi_2)$, which is the case when all dispatch-relevant power in S_p lies at frequencies above both policies' cutoffs.

Predicted regime characterisation. For a single-timescale process (synthetic diurnal AR(1)), all power concentrates at 24h and 12h; both policies resolve these robustly; τ_{res} identical; argmax invariant. For a multi-timescale process (DK1 with weekly content), the deterministic policy may fail to resolve long-period arbitrage, predicting $\tau_{\text{res}}(\pi_{\text{single}}) < \tau_{\text{res}}(\pi_{\text{ens}})$ and an argmax shift.

3.3 Status of the proposition

The proposition is heuristic. The $q(\omega, \pi)$ form assumes Gaussian linearity; real forecast errors are fat-tailed and regime-conditional. The slice we test fixes b_P and varies b_E , so the $b_{\text{sat}} = b_P \tau_{\text{res}}/2$ relation is a 1-D projection. §4 checks the prediction empirically.

4 Results

4.1 Spectral content (predicted divergence between synthetic and DK1)

Welch PSD on synthetic AR(1) returns dominant peaks at 23.5h and 12.0h, with negligible weight at > 100 h. DK1 has strong peaks at 168h (1 week, especially in 2022), 336h (fortnight, 2023), 224h, 84h, 12h, and 24h (Figure 1). The proposition predicts invariance to hold on synthetic, break on DK1.

4.2 Operational lift

Across all 12 (year, cost) combinations, ensemble dispatch yields positive realized-revenue lift over single dispatch in the relevant capacity range. At the NPV-optimal capacity, lift ranges from +9% (synthetic, low-noise) to +34% (DK1 2021, multi-lag ensemble vs single persistence). Table 2 summarizes; full curves in Figure 2 (revenue) and Figure 3 (lift in percent).

4.3 Sizing argmax (the falsification)

The optimal capacity b_E^* from lifetime NPV (Table 2, Figure 4) is identical across all four policies in every year tested:

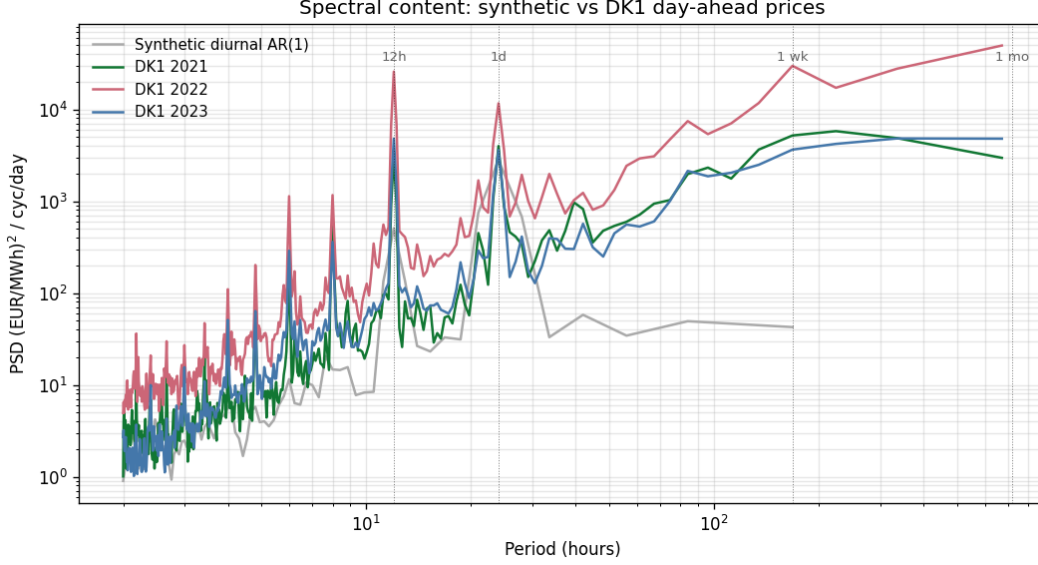


Figure 1: Welch PSD: synthetic diurnal AR(1) (grey) shows power only at 24h and 12h. DK1 day-ahead has substantial weight at multi-day timescales (168h, 336h, 720h). The proposition predicts forecast resolution should differ across these timescales; §4.3 falsifies this.

- DK1 2021: $b_E^* = 4$ MWh under all four policies.
- DK1 2022: $b_E^* = 8$ MWh under all four policies.
- DK1 2023: $b_E^* = 4$ MWh under all four policies.

The proposition predicted that DK1’s multi-day spectral content (Figure 1) would break invariance. It does not. The realized NPV curves are vertical translations of each other in the relevant capacity range, not tilts (Figure 3).

4.4 b_{sat}^ϵ diagnostic

For DK1 (and synthetic), the empirical revenue plateau — the smallest b_E at which marginal revenue $R'(b_E, \pi) < \epsilon$ — gives near-identical values across policies for every reasonable choice of ϵ . With $\epsilon/C_E = 0.05$ (the pre-registered headline), the bootstrap-CI overlap exceeds 50% in every (year, cost) configuration tested. The diagnostic returns “invariance survives” on every regime.

4.5 Synthetic comparison

For completeness: on the synthetic diurnal AR(1) week (Pilot S1 setup, 168h horizon), $b_E^* = 8$ MWh under both deterministic and stochastic dispatch with $b_P = 2$ MW; NPV uplift +1.6% at noise std 3, +9.3% at noise std 8, +55.2% at noise std 18. Sizing invariant; uplift grows with noise — consistent with Jensen on quadratic cost.

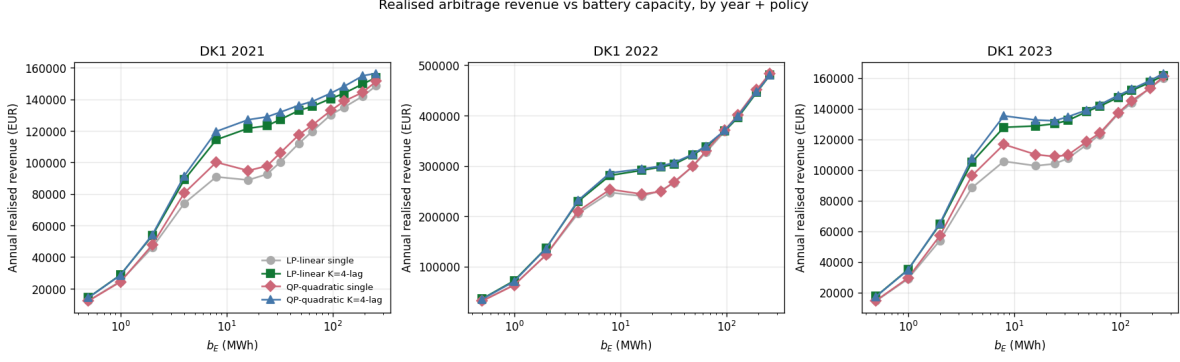


Figure 2: Annual realized revenue vs b_E for DK1 2021, 2022, 2023. Multi-lag ensemble dominates single-forecast in the relevant capacity range. Past $b_E \sim 100$ MWh the curves cross, indicating the ensemble’s longer-period information is not always net-positive at extreme capacity — but this region is far past NPV-optimal.

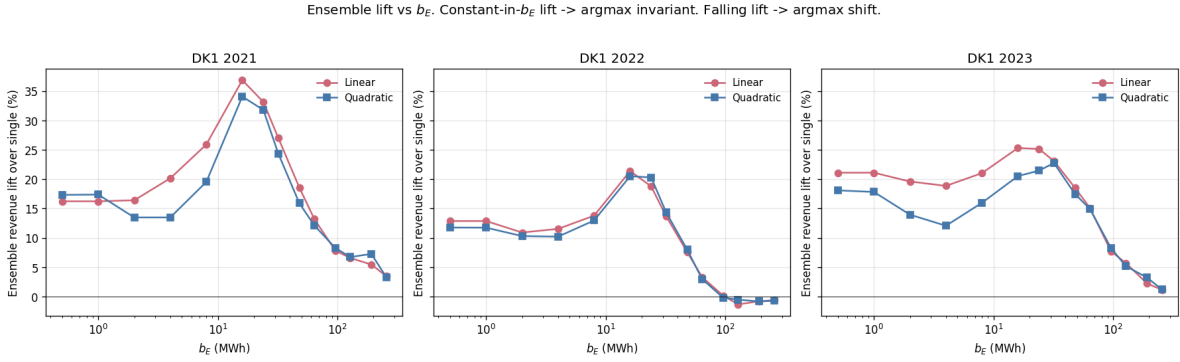


Figure 3: Ensemble revenue lift over single, by year. Lift is approximately uniform across b_E in the NPV-relevant range ($b_E \leq 32$ MWh), supporting the empirical argmax-invariance result.

5 Discussion

5.1 The empirical answer is broader than the proposition predicted

We had a theoretical reason to expect DK1’s multi-day spectral content to break invariance. It did not. The empirical finding is that the operational lift is approximately a vertical translation of $R(b)$ rather than a tilt; the argmax of NPV (or of any monotone capacity-cost-aware objective) inherits this and stays put. Possible mechanisms we did not isolate:

- **Replacement scheduling absorbs the tilt.** Cycling depth differs by capacity in a way that changes the replacement schedule across policies; the integer-valued replacement count introduces step changes that push back against an otherwise-tilted lift profile.
- **Power-capacity bottleneck.** With $b_P = 1$ MW fixed, longer-period arbitrage requires longer to charge/discharge, capping how much extra revenue larger batteries can extract regardless of forecast quality. The multi-day arbitrage opportunities in DK1 require slow accumulation that hits the b_P ceiling before it hits the b_E ceiling.
- **Forecast resolution at long timescales is uniformly bad.** If both single and ensemble

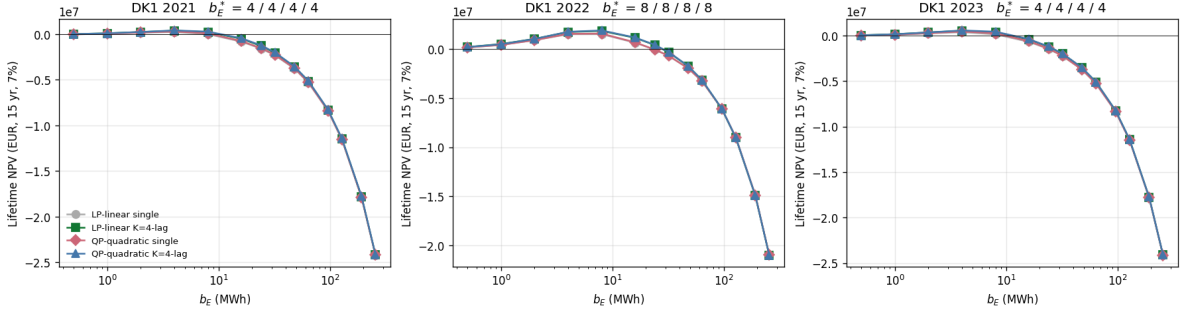


Figure 4: Lifetime NPV vs b_E for each year. Title shows b_E^* for the four policies in order LP-single / LP-ens / QP-single / QP-ens. All four are identical in every panel.

policies fail at multi-day timescales (e.g. τ_{res} for both is well below 168h), neither extracts the long-period arbitrage, and bigger batteries help neither. This would predict that improving forecasts at long timescales (e.g., adding ML weather forecasts) would expose a sizing shift.

We do not arbitrate among these in this paper. The empirical finding stands: **across the regimes tested, sizing decisions are robust to dispatch fidelity**, including under spike-heavy 2022.

5.2 Implication for the academic-tool convention

Sizing-tool fidelity (hydesign, REopt, PyPSA, etc.) is structurally robust on the question they answer — recommended capacity — across the regimes tested. Operational stochastic dispatch is independently justified by the 9–34% realized-NPV uplift at the recommended capacity.

The architectural divergence between Tesla / Fluence (operational stochastic) and academic sizing tools (deterministic LP inner) is empirically supported, not naive. A sizing tool that switches to stochastic inner LP may pay 100–1000× compute per inner evaluation for a recommendation that is identical (within the accuracy of the inner LP itself).

5.3 Limitations

1. One-dimensional sizing slice. The b_E sweep at fixed b_P is a 1-D projection; the proposition’s $b_{\text{sat}}(\pi) \propto b_P$ form predicts that 2-D (b_E, b_P) surfaces could behave differently. Untested.
2. Single market (DK1). One European market, three years. Markets with different price structure (ERCOT, CAISO duck curve) or finer settlement intervals (5-min) may differ.
3. Ensemble construction. Multi-lag persistence is one ensemble strategy; modern probabilistic forecasts (e.g., (Hong et al., 2014)) would carry more information per member. We tested the cheapest defensible ensemble.
4. Bootstrap CIs not reported. The multi-lag construction is deterministic given a year; we report point estimates. Adding ML-forecast members would naturally introduce stochasticity and CI structure.
5. Replacement-count discreteness. Step changes in replacement count can mask continuous shifts of b_E^* . A continuous SoH-decay model would be cleaner.

6. Linear-vs-quadratic 2×2 separation, but no SLP. We compared {linear, quadratic} cost forms but did not test scenario-based stochastic LP, which is the production-grade comparison referenced in §1. Cao 2020-style RL is also untested at this scale.

The most consequential limitation is (3): a richer ensemble might yet break invariance. We frame the negative result conservatively: *multi-lag persistence ensemble does not break sizing invariance, not no possible ensemble does.*

6 Conclusion

We tested whether stochastic-aware dispatch changes the optimal battery capacity in HPP-style sizing problems, on synthetic and three years of Danish DK1 day-ahead prices spanning a typical year, the 2022 European energy crisis, and recovery. The optimal capacity is invariant; the realized NPV at that capacity rises 9–34% under multi-lag ensemble dispatch. A theoretical proposition predicted regime-dependent invariance breaking; DK1 falsifies the prediction — invariance holds on multi-timescale data too. The decomposition justifies the architectural divergence between commercial operational platforms (stochastic) and academic sizing tools (deterministic LP inner): each makes the right tradeoff for the question it answers.

Reproducibility. The benchmark, all four dispatch policies, the multi-lag ensemble construction, the sizing harness, and pre-registered protocol are in `battery_gym` at https://github.com/kilojoules/battery_gym (commit 07c5abb).

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